

Trends in Restaurant Sales and Menu Profitability for Power BI-Based Operational Optimization

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1. Abstract and Introduction

In today's highly competitive market, leveraging data analytics is essential for sustaining business growth and maintaining a competitive advantage. This report presents a comprehensive data visualization and analytical study of a restaurant's sales performance using Microsoft Power BI. The primary objective of this analysis is to address operational inefficiencies by transforming raw transactional data into meaningful and interactive visual insights.

A relevant Restaurant Sales dataset was sourced and processed using Power Query within Power BI. The data was cleaned and transformed to ensure accuracy and consistency before analysis. The study focuses on examining transaction-level data to uncover patterns in customer purchasing behavior, identify peak dining hours, evaluate menu item profitability, and assess overall revenue performance.

Key performance indicators (KPIs), including total revenue, hourly sales trends, category performance, and high-margin menu items, were visualized through interactive dashboards. By analyzing these metrics, the report provides data-driven insights to support managerial decision-making. The findings enable management to optimize staff allocation during peak hours, strategically restructure the menu to promote profitable items, and enhance operational efficiency. Ultimately, this study demonstrates how business intelligence tools such as Power BI can transform raw data into actionable strategies that improve overall business performance and profitability.

2. Problem Statement

The restaurant business has significant operating costs, especially when it comes to labor allocation and inventory management (food waste). The inability to precisely determine peak operating hours and the lack of insight into which menu items (or buffet tiers) generate the highest revenue vs those that perform poorly are the main issues this analysis attempts to solve. Without this information, the restaurant finds it difficult to condense its menu and manage staff scheduling, which lowers overall profitability.

3. Dataset Description

The dataset utilized for this Business Intelligence analysis is a transactional Point of Sale (POS) record sourced from Kaggle. It represents the daily operational and sales activities of a restaurant/retail business. The raw dataset is structured in a CSV format and contains a comprehensive log of customer orders, product details, financial metrics, and operational metadata.

Below is the detailed data dictionary explaining each attribute (column) used in the Power BI data model:

3.1 Transaction & Time Metrics:

order id (Data Type: Text/Nominal): A unique alphanumeric identifier assigned to each individual customer transaction. This is crucial for counting the total number of distinct orders and calculating the average order value (AOV).

date (Data Type: Date): The specific date on which the transaction occurred. This temporal data is essential for time-series analysis, allowing the business to track daily, monthly, and seasonal revenue trends.

3.2 Product & Financial Metrics:

product (Data Type: Text/Categorical): The specific name of the item or menu purchased by the customer. Used primarily for product performance and menu engineering analysis.

price (Data Type: Decimal Number/Quantitative): The unit price of the individual product.

quantity (Data Type: Whole Number/Quantitative): The number of units purchased in a single transaction. (Note: Combined with 'price', this is used to calculate the 'Total Revenue' measure).

purchase (Data Type: Text/Categorical): The category or classification of the item sold (Food, Beverage, or transaction type like Dine-in/Takeaway). This helps in understanding macro-level sales distributions.

payment method (Data Type: Text/Categorical): The medium of exchange utilized by the customer to settle the bill (e.g., Cash, Credit Card, E-wallet). Essential for analyzing customer payment behavior and optimizing checkout operations.

3.3 Operational & Geographical Metrics:

manager (Data Type: Text/Categorical): The name of the shift manager or branch manager responsible at the time of the transaction. This attribute is used to evaluate staff performance and operational efficiency.

city (Data Type: Text/Geographical): The specific city or location where the restaurant branch is situated. This allows for geographical performance mapping and identifying the most and least profitable branches.

4. Data Preprocessing & Methodology

To ensure data accuracy and reliability, the ETL (Extract, Transform, Load) process was performed using Power Query Editor within Power BI.

4.1 Data Cleaning & Formatting: The date column was initially recognized as text due to locale formatting issues. It was transformed into a standardized Date data type using the "Using Locale" function to enable time-series analysis.

	A	B	C	D	E	F
10	10470	11/11/2022	Burgers	12.99	594.27	In-store
19	10471	11/11/2022	Chicken Se	9.95	201.01	In-store
20	10472	11/11/2022	Fries	3.49	630.37	In-store
21	10473	11/11/2022	Beverages	2.95	677.97	In-store
22	10474	11/11/2022	Sides & Otl	4.99	200.4	In-store
23	10475	12/11/2022	Burgers	12.99	523.48	In-store
24	10476	12/11/2022	Chicken Se	9.95	201.01	In-store
25	10477	12/11/2022	Fries	3.49	630.37	In-store
26	10478	12/11/2022	Beverages	2.95	677.97	In-store
27	10479	12/11/2022	Sides & Otl	4.99	200.4	In-store
28	10480	13-11-2022	Burgers	12.99	508.08	In-store
29	10481	13-11-2022	Chicken Se	9.95	201.01	In-store
30	10482	13-11-2022	Fries	25.5	630.37	In-store
31	10486	14-11-2022	Chicken Se	29.05	201.01	In-store
32	10487	14-11-2022	Fries	3.49	630.37	In-store
33	10488	14-11-2022	Beverages	2.95	677.97	In-store
34	10489	14-11-2022	Sides & Otl	4.99	200.4	In-store
35	10490	15-11-2022	Burgers	12.99	508.08	In-store
36	10491	15-11-2022	Chicken Se	9.95	201.01	In-store
37	10492	15-11-2022	Fries	3.49	573.07	In-store
38	10493	15-11-2022	Beverages	2.95	677.97	In-store
39	10494	15-11-2022	Sides & Otl	4.99	200.4	In-store
40	10495	16-11-2022	Burgers	12.99	508.08	In-store
41	10496	16-11-2022	Chicken Se	9.95	201.01	In-store
42	10497	16-11-2022	Fries	3.49	573.07	In-store
43	10498	16-11-2022	Beverages	2.95	677.97	Drive-thru
44	10499	16-11-2022	Sides & Otl	4.99	200.4	Drive-thru

Figure 1 Date type before cleaning

4.2 Data Transformation (DAX): A new calculated measure was created to determine the actual sales value per transaction. The DAX *Total Revenue* = $SUMX('9 Sales-Data-Analysis', '9 Sales-Data-Analysis'[Price] * '9 Sales-Data-Analysis'[Quantity])$ was implemented.

4.3 Methodology: The clean data was then modeled in Power BI. A dashboard was developed using descriptive analytics to visualize historical data, allowing users to filter insights dynamically.

5. Dashboard Explanation

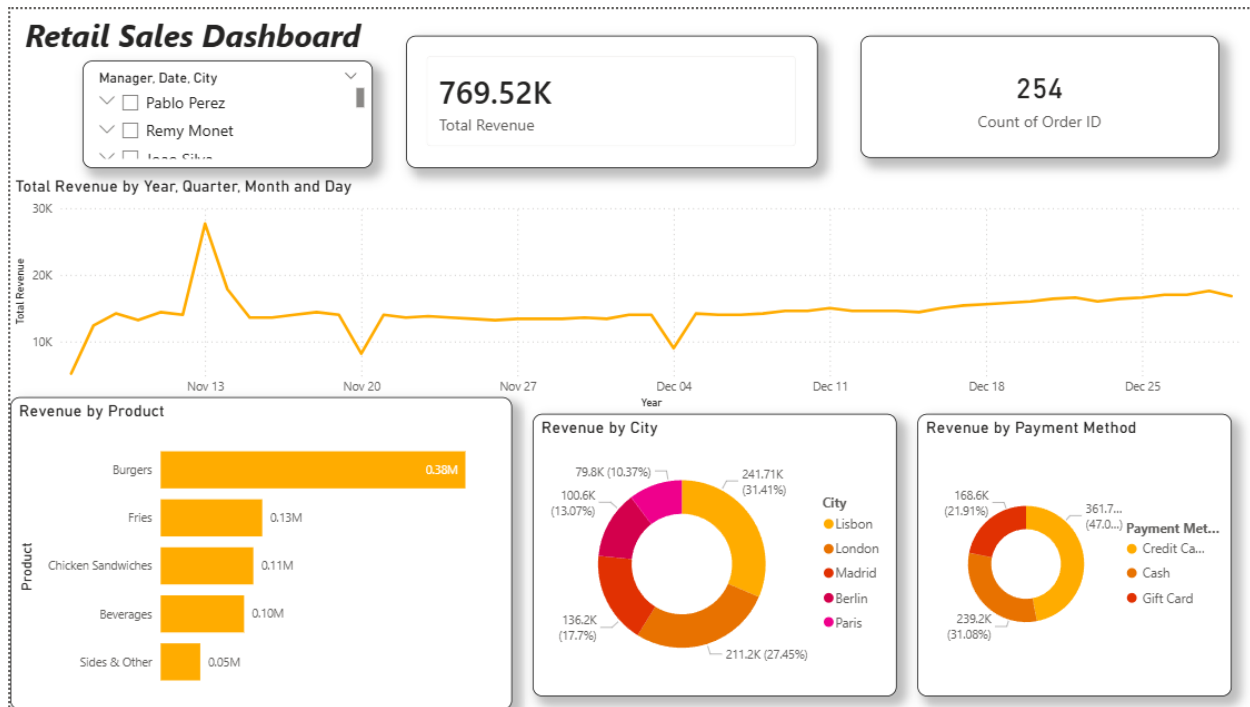


Figure 2 Retail Sales Dashboard

The interactive Power BI dashboard consists of the following key visual elements:

5.1 Slicers (Filters): Interactive filters for date, city, and managers allow users to drill down into specific data subsets.

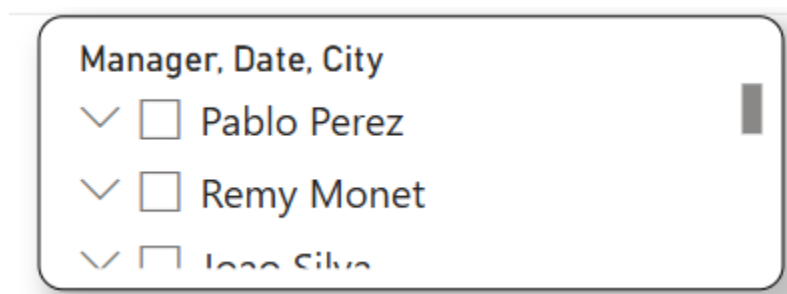


Figure 3 Slicers (Filters)

5.2 Line Chart (Revenue Trend): Displays the Total Revenue over date (X-axis) to highlight seasonal peaks and sales trends.

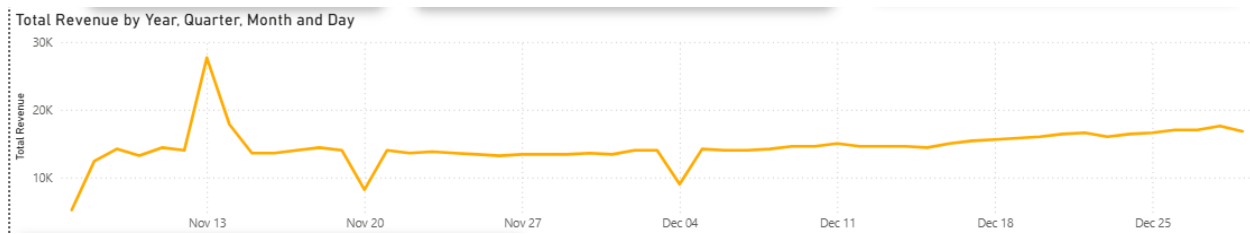


Figure 4 Line Chart

5.3 Clustered Bar Chart (Product Performance): Ranks the product based on Total Revenue, clearly distinguishing top-sellers from low-performers.

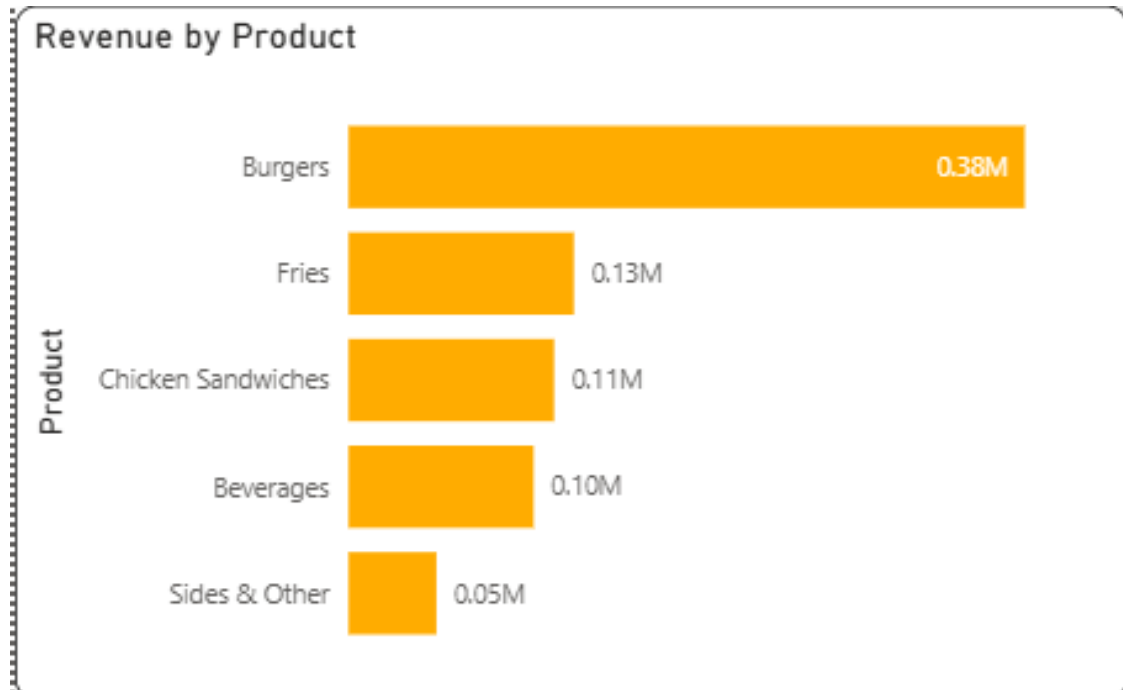


Figure 5 Clustered Bar Chart

5.4 Donut Chart (Sales by City / Payment Method): Illustrates the proportion of total revenue contributed by each city or the percentage share of each payment method.

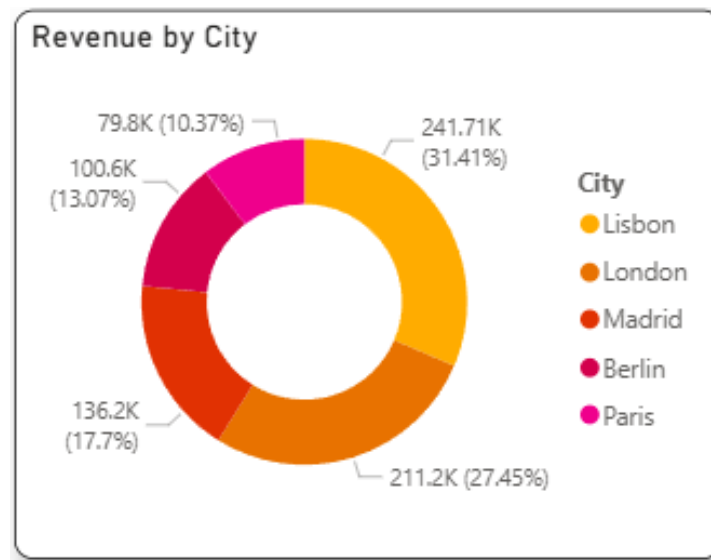


Figure 6 Donut Chart Revenue by City

5.5 Donut Chart (Revenue by Payment Method): Illustrates the percentage share of total revenue generated through different payment methods (Cash, Credit Card, Gift card). This visual provides a clear breakdown of customer payment preferences, which is essential for optimizing checkout operations and financial reconciliation.

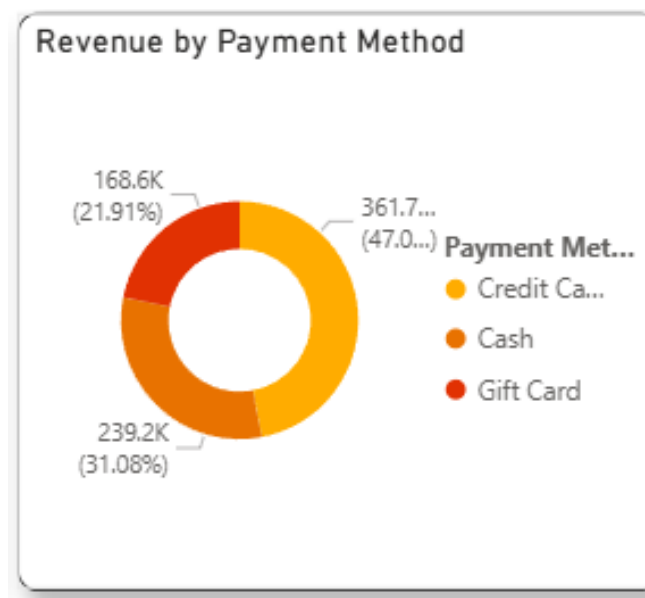


Figure 7 Donut Chart Revenue by Payment Method

6. Analysis and Insights

The visual dashboard revealed a number of important findings, including:

Trend Analysis: According to the line chart, sales peak on Qtr 4 November, indicating periods with more customer foot traffic.

6.1 Product Highlights: The bar chart shows that while "Sides and Other" contributes the least to sales, "Burgers" is the main driver, suggesting that the menu and inventory may need to be revised.

6.2 Operational Insights: The Lisbon branch makes the most money, according to the Donut chart. Furthermore, a sizable majority of consumers favor making purchases with credit cards.

6.3 Customer Payment Preferences: According to the doughnut chart, credit cards account for roughly 47.01% of all sales, which is the majority of the restaurant's revenue. This indicates a strong customer shift towards cashless transactions, whereas Gift card makes up the smallest portion of the revenue.

7. Conclusion

The strengths and limitations of the current sales operations are effectively highlighted by the Power BI analysis. According to the data, a small number of essential items and particular high-performing cities account for a significant portion of revenue.

Suggestions: 1. The company ought to start focused advertising campaigns for its best-selling items. 2. In order to implement best practices to underperforming branches, management should look into the operational tactics of the top-performing manager or city. 3. To boost their turnover rate, low-performing products should be examined for possible elimination or combined with high-performing ones.

8. References (Data Source)

Dataset Name: Restaurant/Retail Sales Dataset

Source: Kaggle

Format: Comma-Separated Values (.csv)

URL: <https://www.kaggle.com/datasets/rohitgrewal/restaurant-sales-data>